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practices

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Author for correspondence:

Daniel W. A. Noble

e-mail: daniel.noble@anu.edu.au

[†]These authors contributed equally to the study.

The promise of community-driven preprints in ecology and evolution

Daniel W. A. Noble¹, Zoe A. Xirocostas², Nicholas C. Wu³, April Robin Martinig^{4,5}, Rafaela A. Almeida⁶, Kevin R. Bairos-Novak⁷, Heikel Balti⁸, Michael G. Bertram^{9,10,11}, Louis Bliard¹², Jack A. Brand⁹, Ilha Byrne⁷, Ying-Chi Chan¹³, Dena Jane Clink¹⁴, Quentin Corbel¹⁵, Ricardo A. Correia¹⁶, Jordann Crawford-Ash¹⁷, Antica Culina¹⁸, Elvira D'Bastiani¹⁹, Gideon G. Deme^{20,21}, Melina de Souza Leite²², Félicie Dhellemmes^{23,24}, Shreya Dimri²⁵, Szymek M. Drobniak^{4,26}, Alexander D. Elsy²⁷, Susan E. Everingham²⁸, Samuel J. L. Gascoigne^{29,30}, Matthew J. Grainger³¹, Gavin C. Hossack³², Knut Anders Hovstad³³, Edward R. Ivimey-Cook³⁴, Matt Lloyd Jones³⁵, Ineta Kačergytė³⁶, Georg Küstner³⁷, Dalton C. Leibold¹, Magdalena M. Mair³⁸, Jake Martin^{9,10,11,39}, Ayumi Mizuno^{4,40}, Iain R. Moodie⁴¹, David Moreau⁴², Rose E. O'Dea⁴³, James A. Orr²⁹, Matthieu Paquet¹⁵, Rabindra Parajuli^{44,45}, Joel L. Pick⁴⁶, Patrice Pottier^{1,4}, Marija Purgar¹⁸, Pablo Recio¹, Dominique G. Roche⁴⁷, Raphaël Royauté⁴⁸, Saeed Shafiei Sabet⁴⁹, Julio M. G. Segovia²⁵, Inês Silva⁵⁰, Alfredo Sánchez-Tójar²⁵, Bruno E. Soares⁵¹, Birgit Szabo⁵², Elina Takola⁵³, Eli S. J. Thore^{9,54}, Bishnu Timilsina⁵⁵, Natalie E. van Dis⁵⁶, Wilco C. E. P. Verberk⁵⁷, Stefan J. G. Vriend⁵⁸, Kristoffer H. Wild^{1,59}, Coralie Williams⁴, Yefeng Yang⁴, Shinichi Nakagawa^{4,60,†} and Malgorzata Lagisz^{4,60,†}

¹Division of Ecology and Evolution, Research School of Biology, The Australian National University, Canberra, Australian Capital Territory 2600, Australia

²School of Life Sciences, University of Technology Sydney, Sydney, New South Wales 2007, Australia

³Hawkesbury Institute for the Environment, Western Sydney University, Richmond, New South Wales 2753, Australia

⁴Evolution and Ecology Research Centre, School of Biological, Earth and Environmental Sciences, University of New South Wales, Sydney, New South Wales 2052, Australia

⁵The Okanagan Institute for Biodiversity, Resilience, and Ecosystem Services, University of British Columbia, Kelowna, British Columbia Canada

⁶Laboratory of Freshwater Ecology, Evolution and Conservation, KU Leuven, Belgium

⁷School of the Environment, The University of Queensland, Brisbane 4072, Australia

⁸Université Marie et Louis Pasteur, CNRS, Chrono-environnement (UMR 6249), F-25000 Besançon, France

⁹Department of Wildlife, Fish, and Environmental Studies, Swedish University of Agricultural Sciences, Umeå 907 36, Sweden

¹⁰Department of Zoology, Stockholm University, Stockholm 114 18, Sweden

¹¹School of Biological Sciences, Monash University, Melbourne 3800, Australia

¹²Department of Evolutionary Biology and Environmental Studies, University of Zurich, Zurich, Switzerland

¹³Swiss Ornithological Institute, Sempach, Switzerland

¹⁴K. Lisa Yang Center for Conservation Bioacoustics, Cornell Lab of Ornithology, Cornell University, Ithaca, NY, USA

¹⁵Theoretical and Experimental Ecology Station (SETE), UAR2029, CNRS, Moulis, France

¹⁶Biodiversity Unit, University of Turku, Turku 20014, Finland

¹⁷Fenner School of Environment and Society, Australian National University, Canberra, ACT, Australia

¹⁸Ruder Bošković Institute, Zagreb, Croatia

¹⁹Department of Ecology and Evolutionary Biology, University of California, LA, USA

²⁰Department of Biology, Case Western Reserve University, Cleveland, OH 44106, USA

²¹Department of Science Laboratory Technology, University of Jos, Jos, Nigeria

²²Department of Ecology, University of São Paulo, São Paulo, Brazil

²³Cluster of Excellence 'Science of Intelligence', Technical University of Berlin, Berlin, Germany

²⁴Center for Adaptive Rationality, Max Planck Institute for Human Development, Berlin, Germany

²⁵Department of Evolutionary Biology, Bielefeld University, Bielefeld, Germany

- ²⁶Institute of Environmental Sciences, Jagiellonian University, Krakow, Poland
- ²⁷Department of Environmental Systems Science, ETH Zürich, Zürich, Switzerland
- ²⁸Institute of Plant Sciences and Oeschger Centre for Climate Change Research, University of Bern, Bern
- ²⁹Department of Biology, University of Oxford, Oxford, UK
- ³⁰School of Biological Sciences, University of Aberdeen, Aberdeen, UK
- ³¹Norwegian Institute for Nature Research, Trondheim, Norway
- ³²Port Perry, Ontario, Canada
- ³³SINTEF Ocean, Trondheim, Norway
- ³⁴School of Biodiversity, One Health and Veterinary Medicine, University of Glasgow, Glasgow, UK
- ³⁵European Centre for the Environment and Human Health, University of Exeter Medical School, Penryn, UK
- ³⁶Department of Ecology, Swedish University of Agricultural sciences, Uppsala, Sweden
- ³⁷Department of Animal Ecology and Tropical Biology, Biocenter, University of Würzburg, Würzburg, Germany
- ³⁸Statistical Ecotoxicology, Bayreuth Center of Ecology and Environmental Research (BayCEER), University of Bayreuth, Bayreuth, Germany
- ³⁹School of Life and Environmental Sciences, Deakin University, Geelong 3216, Australia
- ⁴⁰Faculty of Science, Department of Biology, Hokkaido University, Sapporo, Japan
- ⁴¹Department of Biology, Lund University, Lund, Sweden
- ⁴²Centre for Brain Research, University of Auckland, Auckland, New Zealand
- ⁴³School of Agriculture, Food and Ecosystem Sciences, University of Melbourne, Melbourne, Australia
- ⁴⁴Department of Geosciences, Florida Atlantic University, Boca Raton, FL 33486, USA
- ⁴⁵Odum School of Ecology and Center for Geospatial Research, University of Georgia, Athens, GA 30602, USA
- ⁴⁶Institute of Ecology and Evolution, University of Edinburgh, Edinburgh, UK
- ⁴⁷Department of Biology, Carleton University, Ottawa, Canada
- ⁴⁸Université Paris-Saclay, INRAE, AgroParisTech, UMR EcoSys, Palaiseau, France
- ⁴⁹Fisheries Department, Faculty of Natural Resources, University of Guilan, Swomeh Sara P.O. Box 1144, Iran
- ⁵⁰Center for Advanced Systems Understanding (CASUS), Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Görlitz, Germany
- ⁵¹Institute of Environmental Change & Society, University of Regina, Regina, Canada
- ⁵²Division of Behavioural Ecology, University of Bern, Bern, Switzerland
- ⁵³Department of Computational Landscape Ecology, Helmholtz Center for Environmental Research - UFZ, Leipzig, Germany
- ⁵⁴Laboratory of Adaptive Biodynamics, Research Unit of Environmental and Evolutionary Biology, Institute of Life, Earth, and Environment, University of Namur, Namur, Belgium
- ⁵⁵The Arctic University Museum of Norway, The Arctic University of Norway (UiT), Tromsø, Norway
- ⁵⁶Helsinki Institute of Life Sciences, Helsinki University, Helsinki, Finland
- ⁵⁷Department of Ecology, Radboud University Nijmegen, Nijmegen
- ⁵⁸The Netherlands Institute of Ecology (NIOO-KNAW), Wageningen, The Netherlands
- ⁵⁹School of BioSciences, The University of Melbourne, Victoria 3010, Australia
- ⁶⁰Department of Biological Sciences, University of Alberta, Edmonton T6G 2E9, Canada
- id** DWAN, 0000-0001-9460-8743; ZAX, 0000-0001-7103-5153; NCW, 0000-0002-7130-1279; ARM, 0000-0002-0972-6903; RAA, 0000-0002-5228-9091; HB, 0009-0006-2283-6767; MGB, 0000-0001-5320-8444; LB, 0000-0002-2349-8513; IB, 0000-0003-3909-2902; Y-CC, 0000-0002-7183-4411; QC, 0000-0002-6416-5651; RAC, 0000-0001-7359-9091; ED, 0000-0002-8615-2763; GGD, 0000-0002-0537-6707; FD, 0000-0002-2043-4653; SD, 0009-0009-2155-7506; SJLG, 0000-0002-2984-1810; GCH, 0000-0002-8407-0289; KAH, 0000-0002-7108-0787; ERI-C, 0000-0003-4910-0443; IK, 0000-0003-4756-8253; MMM, 0000-0003-0074-6067; IRM, 0000-0003-3416-1198; REO, 0000-0001-8177-5075; JAO, 0000-0002-6531-5623; RP, 0000-0001-8360-0068; JLP, 0000-0002-6295-3742; PP, 0000-0003-2106-6597; PR, 0000-0002-5890-0218; SSS, 0000-0001-5919-2527; AS-T, 0000-0002-2886-0649; BS, 0000-0002-3226-8621; ESJT, 0000-0002-0029-8404; WCEPV, 0000-0002-0691-583X; SJGV, 0000-0002-9006-5988; KHW, 0000-0001-6714-3311; CW, 0000-0003-1312-4953; SN, 0000-0002-7765-5182; ML, 0000-0002-3993-6127

Publishing preprints is quickly becoming commonplace in ecology and evolutionary biology. Preprints can facilitate the rapid sharing of scientific knowledge establishing precedence and enabling feedback from the research community before peer review. Yet, significant barriers to preprint use exist, including language barriers, a lack of understanding about the benefits of preprints and a lack of diversity in the types of research outputs accepted (e.g. reports). Community-driven preprint initiatives can allow a research community to come together to break down these barriers to improve equity and coverage of global knowledge. Here, we explore the first preprints uploaded to *EcoEvoRxiv* ($n = 1216$), a community-driven preprint server for ecologists and evolutionary biologists, to characterize preprint use in ecology, evolution and conservation. Our perspective piece highlights some of the unique initiatives that *EcoEvoRxiv* has taken to break down barriers to scientific publishing by exploring the composition of articles, how gender and career stage influence preprint use, whether preprints are associated with greater open science practices (e.g. code and data sharing) and tracking preprint publication outcomes. Our analysis identifies areas that we still need to improve upon but highlights how community-driven initiatives, such as *EcoEvoRxiv*, can play a crucial role in shaping publishing practices in biology.

1. Introduction

Publishing preprints—papers communicating non-peer-reviewed research findings—is now an entrenched practice across a multitude of scientific disciplines [1]. Preprints in biology have had a slower uptake relative to other disciplines [2], but new discipline-specific preprint servers, such as *EcoEvoRxiv* (<https://ecoevorxiv.org>), provide a means by which ecologists and evolutionary biologists can disseminate research findings. Preprints attempt to break down barriers to scientific publishing by: (i) increasing the visibility of research and the speed at which research findings become available, which can lead to more

citations (e.g. [3,4]); (ii) helping establish the precedence of research findings; (iii) removing financial barriers to open access publication; and (iv) enabling earlier feedback from the research community [5–7]. Ultimately, preprints can facilitate the rapid sharing of scientific knowledge that can have significant impacts on fundamental and applied knowledge globally [8].

Preprint servers can empower researchers to make their research findings more accessible, open and transparent but only if they are used as forums for spreading and discussing findings within a research community. However, significant barriers to the widespread adoption of preprints remain, ranging from a lack of clarity around preprint policies in journals [9] to a stigma within the research community that preprints are of poor quality [10] (but see [11]). Nonetheless, we lack an understanding of the factors that influence preprint use in ecology and evolution. Such an understanding may help improve current initiatives, inform future ones and allow us to work harder in further breaking down barriers to scientific publishing.

EcoEvoRxiv is one of the few community-driven preprint servers that has paved the way for new initiatives, by accepting multilingual preprints, registered reports and non-traditional research reports. Such initiatives are distinct from other preprint servers, such as *bioRxiv*, which only accepts empirical research in English. In addition, community-driven servers like *EcoEvoRxiv* aggregate papers presenting research on similar topics, improving discoverability and opportunities for within-community debate compared with broader preprint servers. *EcoEvoRxiv* promotes peer review and community discussion in the hope of improving the quality of preprints and speeding up their peer-reviewed publication. For example, we encourage authors to use peer community review services such as Peer Community In (PCI) [12], which allow for fast, constructive peer review around a preprint with peer reviews being transparent and published online [12]. *EcoEvoRxiv* also allows authors to submit both preprints and postprints (also known as author-accepted manuscripts). While preprints are versions of manuscripts posted by authors before peer review, postprints are versions of peer-reviewed and accepted articles but without typesetting and formatting by a journal. The main reason for publishing postprints on a preprint server is to ensure published articles are openly accessible to everyone without a paywall (i.e. green open access). Even for articles published open access, depositing the postprint in a repository (e.g. Zenodo) or a preprint server strengthens permanence and access to the content of the article in the event of a journal's collapse or disappearance. Postprints can be published anytime if journals allow it (which many do; see <https://www.sherpa.ac.uk/romeo/>).

Here, we explore the first preprints/postprints ($n = 1216$) uploaded to *EcoEvoRxiv* to characterize preprint practices in ecology and evolution. We aim to understand: (i) in what countries authors who use *EcoEvoRxiv* are located; (ii) the taxonomic diversity of study systems used across articles; (iii) whether preprint server use depends on career stage and gender; (iv) the extent to which authors make use of preprint servers for reports and community-driven peer review; (v) the extent to which data and code are shared in preprints; and (vi) how many preprints remain unpublished, and for those that are published, how long it took for them to become published. In the process, we also provide a summary of what makes *EcoEvoRxiv* distinct from other preprint servers to help further clarify the benefits of using community-driven preprint servers to disseminate research findings.

2. Getting to know your *EcoEvoRxiv* preprint server

EcoEvoRxiv is run by the Society for Open, Reliable and Transparent Ecology and Evolutionary Biology (SORTEE) [13]. Originally launched in 2018 on the Center for Open Science preprint platform, *EcoEvoRxiv* has become a popular preprint server for ecologists and evolutionary biologists. The server has since been adopted by the California Digital Library (CDL). Editors are ecologists and evolutionary biologists from across the globe who volunteer their time to screen submitted papers and push new initiatives in the preprint space. To better understand preprint (and postprint) use on *EcoEvoRxiv*, we downloaded metadata on the accepted articles available on *EcoEvoRxiv* as of 30 September 2023 (see electronic supplementary material for more details on methods). We consider both preprints and postprints as 'articles'. After removing five duplicate titles (suggesting that a few authors uploaded their articles as separate submissions rather than updating the existing article), we extracted data for a total of 1216 articles over the last two years (figure 1A). For more details on the data collection process, see the electronic supplementary material (https://daniel1noble.github.io/ecoevo_1000/).

(a) Overview of *EcoEvoRxiv* preprints (and postprints)

EcoEvoRxiv hosts articles from authors based in 56 countries, with 90% coming from just 17 countries. North America, Australia and European countries upload the most preprints, with many fewer coming from countries in Africa, Central America and parts of Asia (figure 1B). Articles cover all major taxonomic groups, with the most common groups being vertebrates (47.2%), plants (21.9%) and invertebrates (17.5%) (figure 1C).

(b) Diversifying article types on *EcoEvoRxiv*: overcoming the 'grey literature' problem

Accepting a greater diversity of article types allows *EcoEvoRxiv* to help deal with the 'grey literature' problem, whereby data that are relevant for research syntheses are not published in typical peer-reviewed journals [14,15]. *EcoEvoRxiv* has made a concerted effort to diversify the types of articles hosted. This is reflected by 6.2% of the articles being books, book chapters, reports and other research output types, which are typically considered 'grey literature' in ecology and evolutionary biology. As a result, articles on *EcoEvoRxiv* are more diverse than those on other preprint servers that have more restrictive submission policies. For example, *bioRxiv* only accepts empirical research articles (<https://www.biorxiv.org/submit-a-manuscript>).

Empirical research articles are still the most common type of articles submitted to *EcoEvoRxiv* (46.3%), followed by reviews and meta-analyses (26.5%) and opinion papers (10.6%) (figure 1D). Currently, *EcoEvoRxiv* does not host many reports,

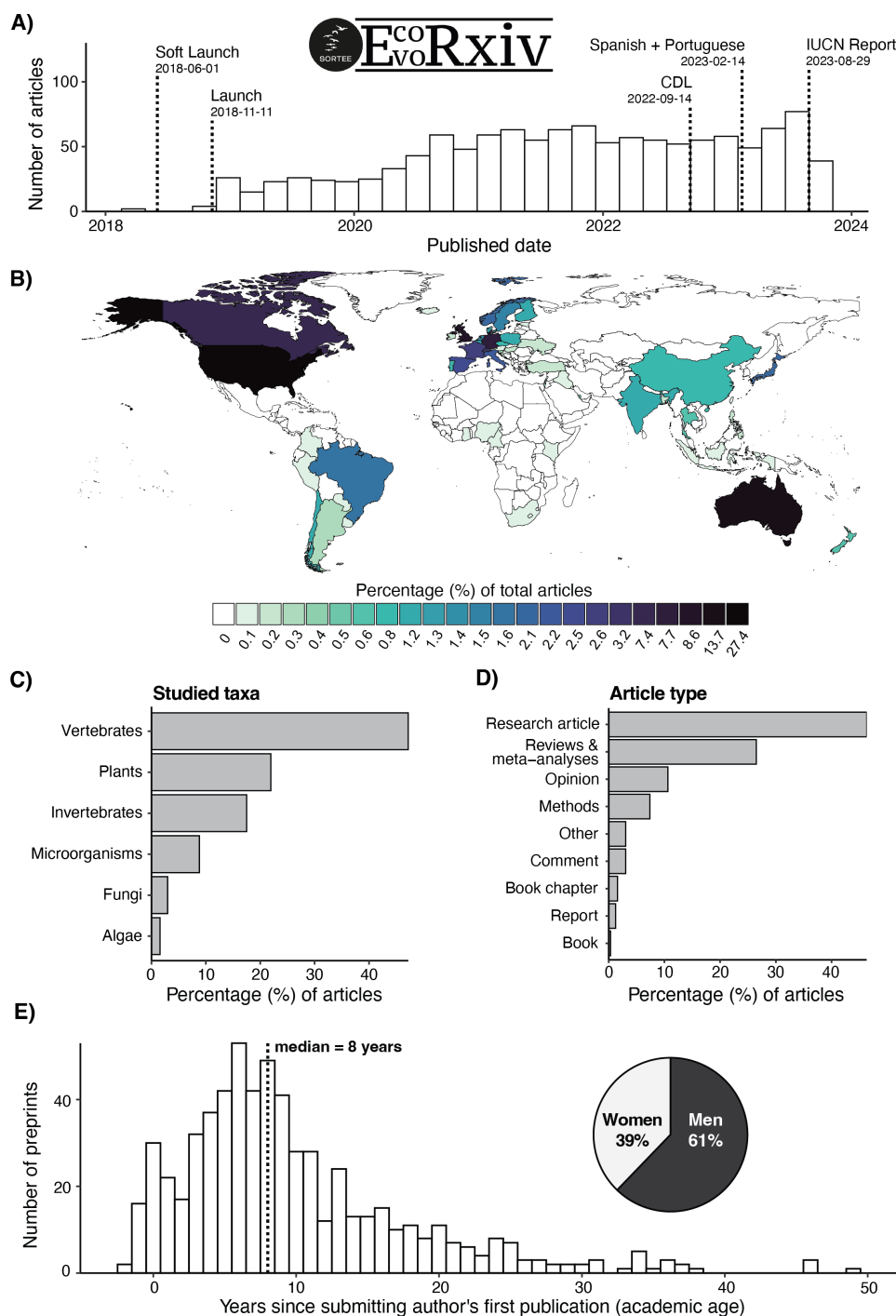


Figure 1. Summary of articles posted to *EcoEvoRxiv*. (A) Number of articles (preprints and postprints) published in *EcoEvoRxiv* between 2018 and 2023. *EcoEvoRxiv* was established in June 2018 before the launch in November 2018. Notable milestones include *EcoEvoRxiv* transitioning to the California Digital Library (CDL), the acceptance of preprints and postprints in Spanish and Portuguese and the acceptance of the first IUCN Red List Ecosystem report. (B) Geographic origin of articles uploaded to *EcoEvoRxiv*, inferred from the country of affiliation of the submitting author. (C) Taxa covered in the articles posted to *EcoEvoRxiv* ($n = 1080$ articles covering relevant taxa). (D) Types of articles accepted on *EcoEvoRxiv* ($n = 1216$ articles). (E) academic age of authors posting preprints to *EcoEvoRxiv* ($n = 1135$ published and unpublished preprints) along with the gender of the submitting author. Values lower than zero are indicative of authors who uploaded preprints before their first scientific publication in a journal. Map base source: R package maps v. 3.4.2. Shapefile: Natural Earth <https://www.naturalearthdata.com/about/terms-of-use/>.

particularly from government or industry, but has formed fruitful partnerships with the International Union for Conservation of Nature (IUCN). For example, IUCN Red List Ecosystem Reports are now posted in *EcoEvoRxiv*, and our community has been able to work closely with the IUCN to ensure these documents meet the IUCN requirements. We encourage authors to consider posting books, book chapters and reports to ensure that they are openly accessible and more easily found. Accepted *EcoEvoRxiv* submissions are given a unique DOI and are indexed on Google Scholar. DOIs can be used in grant applications, CVs and other documents to provide a link to the work.

(c) Breaking down language barriers to scientific communication: improving diversity and data representation globally

A significant barrier to the communication of research findings is the fact that they are primarily communicated in English [16–18]. Research communication through a single language has major consequences for the global distribution of knowledge, resulting in knowledge gaps across some of the most biodiverse and threatened regions in the world [19,20]. Such gaps also impact research syntheses and meta-analyses because they create a distorted picture of our knowledge base that can affect future research, policy development and decision-making [20–23].

EcoEvoRxiv is the only preprint server to date that breaks down language barriers to scientific communication by accepting not only English but also Spanish-, Portuguese- and French-language articles. *EcoEvoRxiv* plans to expand to other languages as new non-English editors for different languages become available. Such initiatives are incredibly important if we are to begin filling global gaps in scientific knowledge. However, multilingual initiatives have been slow to take off on *EcoEvoRxiv*, with only a few Spanish submissions, and a single Portuguese article posted since *EcoEvoRxiv* began accepting non-English articles in 2023. Part of the challenge in getting authors to submit articles in non-English languages is a lack of awareness of *EcoEvoRxiv* in non-English-speaking countries, cultural differences in the perception of preprints and a strong reliance on traditional publishing models that typically mandate publishing in English [24].

(d) Generational and gender-based gaps in preprinting practices

Research can take a while to be published. Early and mid-career researchers (EMCRs) (approximately 10 years post-PhD) are under pressure to publish rapidly to be competitive in job applications, promotions and obtaining grants to progress their careers [7,25]. Preprints are especially useful for EMCRs because they can achieve faster dissemination and greater visibility [4]. EMCRs may therefore be expected to make use of preprints more than colleagues at later career stages because they are more often in charge of article submission and have developed their careers in an environment where preprints are a normal part of the publication process. We collected data on the ‘academic age’ of submitting authors by looking at Google Scholar profiles of authors (when available) and recording their first year of publication in a peer-reviewed journal. While this is a rough estimate of career stage, there was evidence that the number of preprints posted decreases with later career stages (negative binomial glm: year slope = -0.1 , s.e.: 0 , $p < 0.001$, $n = 50$ years). Most preprints were submitted by authors who published their first paper in the last 10 years (figure 1E), with the median year since the first publication being 2013 (mean = 2010.7; s.d. = 9.9, $n = 1133$). These patterns support the expectation that EMCRs may use preprints to make their work more visible and disseminate their findings more quickly. However, we acknowledge that to understand the reasons why EMCRs might adopt preprint servers more readily requires community surveys, as have been done in previous studies (e.g. [11]).

Gender differences in preprint use and publication outcomes have also been observed in several research fields, including ecology and evolutionary biology [26,27]. For example, gender gaps in preprint submissions were observed during COVID-19 lockdowns [28], and previous surveys have shown that female participants are less likely to suggest posting articles as preprints, suggesting gender differences in views around preprints [11]. Therefore, such discrepancies are expected to manifest in preprint use on *EcoEvoRxiv*, but it is unclear to what extent. Understanding gender publishing patterns is challenging with observational data such as ours because we cannot know the gender of authors for certain, but we can use a data-driven approach to ascertain the probability that a particular name is of a given gender (man or woman). We used the R package *gender* v. 0.6.0 [29] to predict the most likely gender of the submitting author of a preprint. We used an algorithm to assign binary gender based on the submitting author’s name. We only used the algorithm-assigned gender when the gender of a given name was identified with 95% certainty. For the remaining names, we performed manual searches to determine gender based on the pronouns and photographs from professional and personal websites. We acknowledge that our approach does not capture self-assigned and non-binary genders. As such, our assumptions about an author’s gender identity may be incorrect. Our data on gender had only two missing values—one where the first name of the submitting author was missing and the other one for a collective submission. As expected, we found that women were less likely to post to *EcoEvoRxiv* compared with men (women: 38.5%; figure 1E), reinforcing existing disparities between male and female scientists. For example, studies have shown that female first authors have lower acceptance rates and are cited less (approx. 2%) compared with males (e.g. [26]).

3. Following the journey of a preprint on *EcoEvoRxiv*: from submission to publication

(a) Science takes time, but publication can take longer

Increased competition in science has raised the bar with respect to the amount of data required for publication [7]. This requirement is a good outcome if it results in higher impact research that better clarifies our understanding of the natural world, but it does come at a cost for the speed of research dissemination [7]. Long publication times can adversely affect EMCRs who rely on publications for job applications, promotions and obtaining grants. Getting research out quickly can also be critical for the development of new knowledge that can reshape research landscapes, which was important during the COVID-19 pandemic [30]. Preprints have been proposed as a way to disseminate research more quickly as it can take a long time before results are ultimately published after formal peer review [6,7]. However, data on the time to publication are needed to quantify the real benefit of preprints in this context.

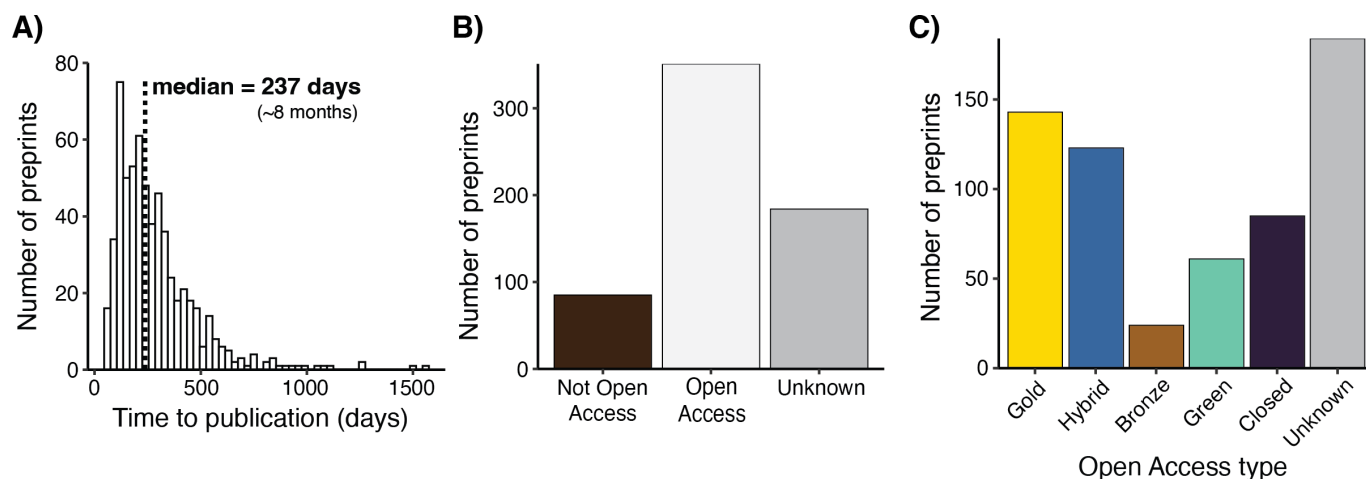


Figure 2. Summary of the publication status of preprints on *EcoEvoRxiv*. (A) The time between uploading a preprint to *EcoEvoRxiv* and its publication as a peer-reviewed journal article. Preprints were considered those articles that were published in a journal a minimum of two months (60 days) after being posted in *EcoEvoRxiv*. (B) Access status of published preprints on *EcoEvoRxiv* classified as ‘open access’ or ‘not open access’. ‘unknown’ status is for articles whose status was unclear on the unpaywall platform. (C) Sub-types of open access status of published preprints on *EcoEvoRxiv*. Sub-type meanings are as follows: ‘green’, articles published in ‘toll-access journals but achieved in an open access repository; ‘bronze’, articles are free to read on publishers’ websites without a license but grant no other rights and can be delayed free-to-read; ‘hybrid’, articles are free to read upon publication with an open access license; ‘gold’, articles published in fully open access journals. For full details on the meaning of each category, see <https://support.unpaywall.org/support/solutions/articles/44001777288-what-do-the-types-of-oa-status-green-gold-hybrid-and-bronze-mean->

We estimated how long it takes to publish a preprint in ecology and evolution by recording the time between when a preprint was first posted on *EcoEvoRxiv*, and its final acceptance in a peer-reviewed journal. In total, 515 preprints remained unpublished (45.4%, $n = 1135$) at the time when these data were collected. Not all of these preprints, however, are anticipated to be published in a peer-reviewed journal (e.g. reports). Nonetheless, the median time to publication for preprints was 237 days (eight months) for all preprints that ended up being published (mean = 286.4; s.d. = 193.9 days) with the maximum time to publication being 1549 days or 4.2 years (figure 2A). For a full breakdown of the time to publication based on article type, refer to electronic supplementary material, table S1. Our results largely confirm the extended timeframes that most authors experience between writing their research papers and their publication.

(b) Cautious ‘open’-mindedness of research in preprints

In addition to speeding up dissemination, preprints and postprints can also be a useful way to ensure that research remains open and accessible to the research community irrespective of the accessibility of the final peer-reviewed paper [6,7]. We evaluated whether preprints and postprints hosted at *EcoEvoRxiv* and that were also published in a journal were published open access. The open access status of each published article was obtained using the R package *roadoi* (v. 0.7.2) to connect to the unpaywall platform [31]. Most of the published preprints and postprints were open access (80.5% ($n = 351$ out of 436, where the status was known); figure 2B); however, 19.5% ($n = 85$) were published behind paywalls. Published articles behind paywalls may otherwise remain inaccessible if it were not for *EcoEvoRxiv*. For preprints and postprints published in open access journals, the type of open access also varied widely (e.g. gold, hybrid and green OA; figure 2C). Such a result may not be too surprising given that authors using preprint servers are probably already ‘pro-open access’, particularly given that *EcoEvoRxiv* is run by the SORTEE.

Data and code sharing are also key components of open science [32]. In the spirit of ‘openness’, we expected data and code sharing among preprints and postprints to be greater than in many papers published in research journals [32,33]. Despite this, we found that 54.4% ($n = 398$) of articles relying on data (i.e. classified as ‘research articles’) on *EcoEvoRxiv* did not share data, and 58.1% ($n = 425$) did not share code.

Authors may be reluctant to share data and code for preprints because of the perceived concern that others may acquire and use their data and code before publication in a journal. Authors of 28.7% ($n = 123$) preprints did not share data at the preprint stage but ultimately did share data when the article was published, whereas authors of 35.2% ($n = 151$) never shared data and 36.1% ($n = 155$) shared data at both stages. The same was true for code. Overall, 16.8% ($n = 72$) of preprints had no open code at the preprint stage, but ultimately did share code at the published article stage. In contrast, authors of 45.2% ($n = 194$) preprints did not share code at either stage while 38% ($n = 163$) shared code at both stages. Relatively low code and data-sharing practices in our sample are consistent with analyses of sharing practices for published articles (e.g. [33]), even for journals with strict public data archiving policies [32].

(c) Paving our future to open, transparent and community-driven science

Our analysis has allowed us to better understand preprinting/postprinting practices in *EcoEvoRxiv*. Overall, *EcoEvoRxiv* articles are diverse but with primary research articles on vertebrates comprising most of the articles posted. North America, Europe and Australia use *EcoEvoRxiv* the most, with very few non-English-language articles deposited to date. Submitting authors who were earlier in their career and more often with 'male-associated names' tended to use *EcoEvoRxiv* the most. Articles posted to *EcoEvoRxiv* tend to take up to eight months to be published, with many articles not being open access. Code and data sharing were also relatively uncommon at the preprint stage. We attempted to collect data on community discussion around preprints. However, no such data were found on preprint landing pages, likely reflecting inadequate functionality and cross-linking with sources where such discussion is occurring. Based on the insights from our analysis, we provide recommendations to authors and the scientific community on ways they can further promote open and transparent research through preprints:

- (i) First, share your data and code at the preprint stage. Sharing data and code early can help improve the quality of research, establish precedence and improve the transparency and computational reproducibility of scientific findings [25,34]. Reassuringly, sharing data and code is rarely associated with the 'scooping' of research findings [35]. If authors are worried about data being used unintentionally, clear information surrounding its reuse can be included in a license (see <https://choosealicense.com>). Data can also be archived with an embargo on its reuse [36].
- (ii) Second, take advantage of peer-reviewing services such as PCI. The time between posting a preprint and publication is still quite long (approx. eight months). One possible explanation is that preprints are not being sent to suitable journals or are struggling to get into review, slowing down constructive feedback that can improve the quality of a paper. Using PCI circumvents editorial decisions without review, yet only 1.8% of articles ($n = 20$) used PCI. Using such services will ensure that authors receive faster feedback on a paper. Ninety-three journals currently accept PCI reviews and recommendations when considering a paper for publication (<https://peercommunityin.org/pci-friendly-journals/>).
- (iii) Third, seek out and contribute to constructive feedback on preprints [6]. While it is clear that preprints help establish precedence and allow findings to be openly accessible, it still seems rare that constructive discussions form around preprints in an open forum (e.g. *bioRxiv* [37]). Unfortunately, the *EcoEvoRxiv* website does not provide opportunities for discussion given the limitations of the web server at this point in time. As such, we could not accurately assess how much discourse around a given preprint occurs. One way to facilitate such discussions may be to use open preprint peer-review services such as PCI or *PubPeer* (see also [38]) to provide feedback on preprints. Both *PCI* and *PubPeer* provide opportunities for open peer review around a preprint. Peer discussions are given a unique DOI, which can then be associated with a preprint on *EcoEvoRxiv*. However, *EcoEvoRxiv* currently lacks connection to *PCI*, *PubPeer* and Altmetric data, which would allow for discussion to be assimilated around a preprint in one place and make it easier for readers to follow discourse around a preprint. Clearly, as a community, we need to provide better platforms and workflows that document discussions around preprint findings. Such discussions help authors improve their work and communicate their findings more effectively (when done constructively, of course). The lack of discussion around preprints more generally might also be a function of the time constraints researchers face and the lack of credit received for such community service. An important future goal of *EcoEvoRxiv* is to provide better community discussion forums and integration with existing preprint peer-review services. We also need to find new ways to give credit to colleagues who contribute to community discussion.
- (iv) Finally, keep your preprints updated. While most preprints get seamlessly connected and merged with their published version, some remain 'disconnected' as separate articles. Incorrect cross-linking by indexing platforms (e.g. Google Scholar) can create confusion and lead to frustration among authors. The reasons for unmatched preprints and publications are well understood and usually easily rectified. They often result from a mismatch between preprint and published metadata (e.g. titles and author details). For example, nearly one-third of the articles changed their title from preprint submission to publication (30.5%; $n = 229$). We found that mismatched metadata almost always contributed to preprints and published articles not being matched automatically in Google Scholar. At times, further manual merging by authors is needed to connect the preprint and published article (this can be done in Google Scholar). Regardless, we recommend that authors update their preprints with the publication DOI when accepted to journals, especially if their titles have changed. This is very easy for authors to do on *EcoEvoRxiv*, and it should increase the chances that the preprint is correctly linked to the published article and citations are appropriately merged.

Despite the early successes of the new initiatives taken by *EcoEvoRxiv*, as described above, much work remains to be done to improve the understanding and use of preprints and postprints within our community. We view this perspective piece as a small step towards achieving that goal. We hope that readers are more familiar with the benefits of using community-driven preprint servers and the unique initiatives they can pursue. Community-driven preprint servers can set their own agenda and are driven by the needs and desires of the community. Supporting these initiatives should be a priority for all researchers. Volunteers at *EcoEvoRxiv* are encouraged to remain open to new and innovative ways to improve publication and open science practices. Our analysis can be used to drive changes in *EcoEvoRxiv* to make it a better platform for our community. We believe that the future of preprints is bright, and community-driven initiatives, such as *EcoEvoRxiv*, will play a crucial role in the future of scientific publishing.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. All data and code can be found on GitHub at [39]. It is also available on Zenodo [40].

Declaration of AI use. The authors declare that AI was not used to write or edit the manuscript.

Authors' contributions. D.W.A.N.: conceptualization, data curation, formal analysis, investigation, methodology, project administration, supervision, visualization and writing—original draft; Z.A.X.: formal analysis, investigation, visualization and writing—review and editing; N.C.W.: formal analysis, investigation, visualization and writing—review and editing; A.R.M.: investigation, visualization and writing—review and editing; R.A.A.: investigation, validation and writing—review and editing; K.R.B.-N.: investigation, validation and writing—review and editing; H.B.: investigation, validation and writing—review and editing; M.G.B.: investigation, validation and writing—review and editing; L.B.: investigation, validation and writing—review and editing; J.A.B.: investigation, validation and writing—review and editing; I.B.: investigation, validation and writing—review and editing; Y.-C.C.: investigation, validation and writing—review and editing; D.J.C.: investigation, validation and writing—review and editing; Q.C.: investigation, validation and writing—review and editing; R.A.C.: investigation, validation and writing—review and editing; J.C.-A.: investigation, validation and writing—review and editing; A.C.: investigation, validation and writing—review and editing; E.D.: investigation, validation and writing—review and editing; G.G.D.: investigation, validation and writing—review and editing; M.d.S.L.: investigation, validation and writing—review and editing; F.D.: investigation, validation and writing—review and editing; S.D.: investigation, validation and writing—review and editing; S.M.D.: investigation, validation and writing—review and editing; A.D.E.: investigation, validation and writing—review and editing; S.E.E.: investigation, validation and writing—review and editing; S.J.L.G.: investigation, validation and writing—review and editing; M.J.G.: investigation, validation and writing—review and editing; G.C.H.: investigation, validation and writing—review and editing; K.A.H.: investigation, validation and writing—review and editing; E.R.I.-C.: investigation, validation and writing—review and editing; M.L.J.: investigation, validation and writing—review and editing; I.K.: investigation, validation and writing—review and editing; G.K.: investigation, validation and writing—review and editing; D.C.L.: investigation, validation and writing—review and editing; M.M.M.: investigation, validation and writing—review and editing; J.M.: investigation, validation and writing—review and editing; A.M.: investigation, validation and writing—review and editing; I.R.M.: investigation, validation and writing—review and editing; D.M.: investigation, validation and writing—review and editing; R.E.O.: investigation, validation and writing—review and editing; J.A.O.: investigation, validation and writing—review and editing; M.Pa.: investigation, validation and writing—review and editing; R.P.: investigation, validation and writing—review and editing; J.L.P.: investigation, validation and writing—review and editing; P.P.: investigation, validation and writing—review and editing; M.Pu.: investigation, validation and writing—review and editing; P.R.: investigation, validation and writing—review and editing; D.G.R.: investigation, validation and writing—review and editing; R.R.: investigation, validation and writing—review and editing; S.S.S.: investigation, validation and writing—review and editing; J.M.G.S.: investigation, validation and writing—review and editing; I.S.: investigation, validation and writing—review and editing; A.S.-T.: investigation, software, validation and writing—review and editing; B.E.S.: investigation, validation and writing—review and editing; B.S.: investigation, validation and writing—review and editing; E.T.: investigation, validation and writing—review and editing; E.S.J.T.: investigation, validation and writing—review and editing; B.T.: investigation, validation and writing—original draft; N.E.v.D.: investigation, validation and writing—review and editing; W.C.E.P.V.: investigation, validation and writing—review and editing; S.J.G.V.: investigation, validation and writing—review and editing; K.H.W.: investigation, validation and writing—review and editing; C.W.: investigation, validation and writing—review and editing; Y.Y.: investigation, validation and writing—review and editing; S.N.: conceptualization, data curation, formal analysis, investigation, project administration, supervision, visualization and writing—review and editing; M.L.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, visualization and writing—original draft, writing—review and editing.

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